ASSIGNMENT 07

Colour Blind Mode for the GeM Tractors Website

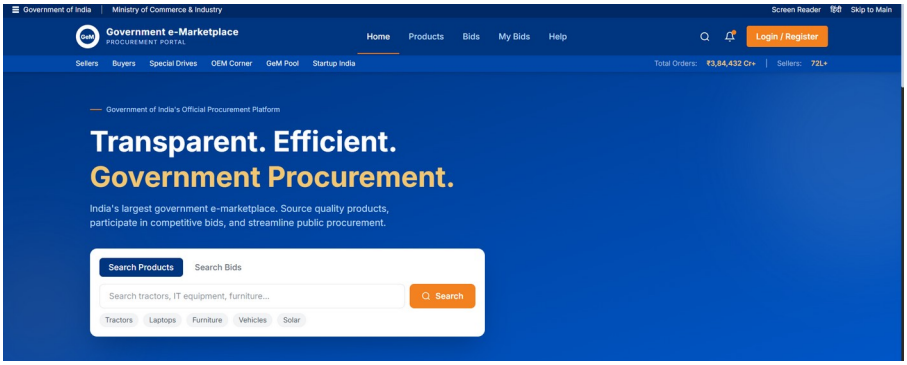
Assignment No.	07
Name	Jasjot Singh
PRN	1032232147
Group	
Assignment Title	Added Colour Blind option to optimize the website for colour-blind people

Objective

To optimize the GeM Tractors website for colour-blind users by adding a Colour Blind Mode option in the header and ensuring that important information (bid deadlines, verification, selection, bid status and actions) is conveyed through text, icons and contrast rather than colour alone.

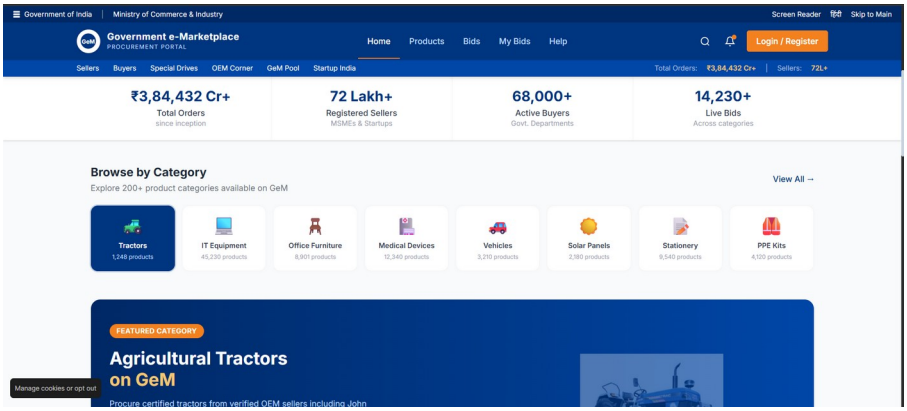
Website Screenshots

Figure 1: Home page – header and hero search



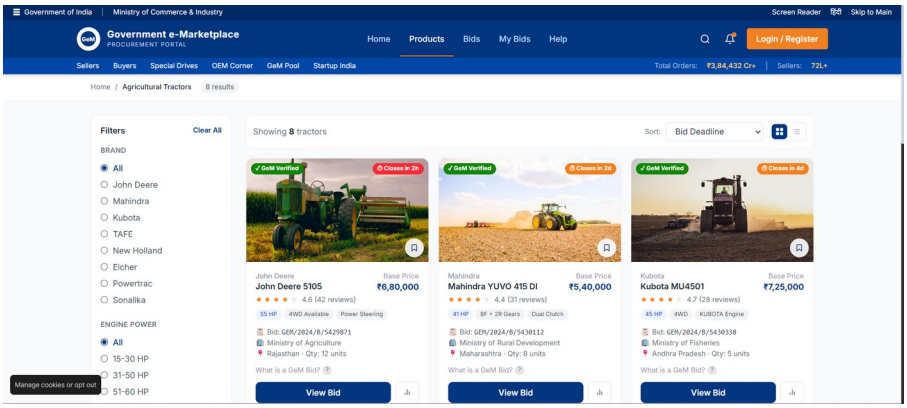
White and gold text on a deep blue background gives strong contrast, and Login / Register, Search Products and Search Bids are all labelled in text.

Figure 2: Browse by Category – Tractors tile



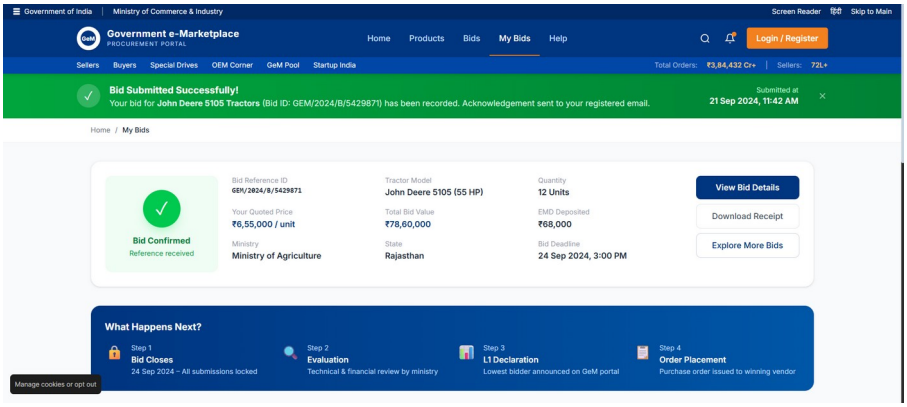
Every category tile carries a text label and product count, and the selected Tractors tile uses a dark fill with white text instead of a colour change alone.

Figure 3: Tractor listing – badges, ratings and filters



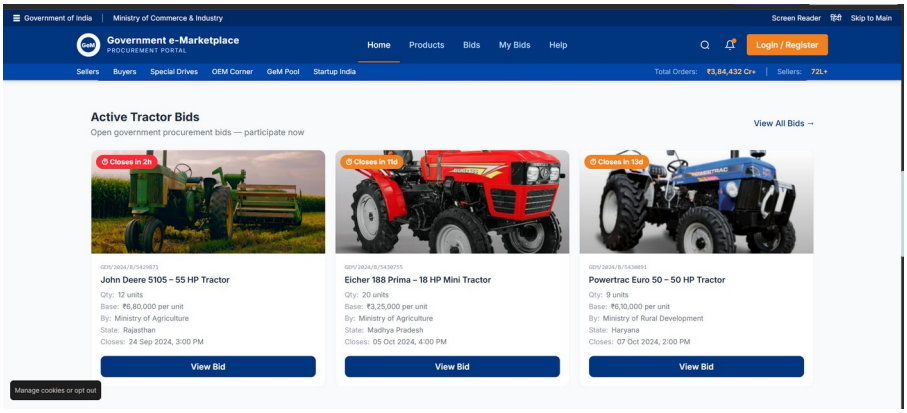
"GeM Verified" pairs a tick with text, "Closes in 2h / 2d / 4d" pairs a clock with the time, and ratings show a number with the review count, so none depends on colour alone.

Figure 4: Bid confirmation – My Bids



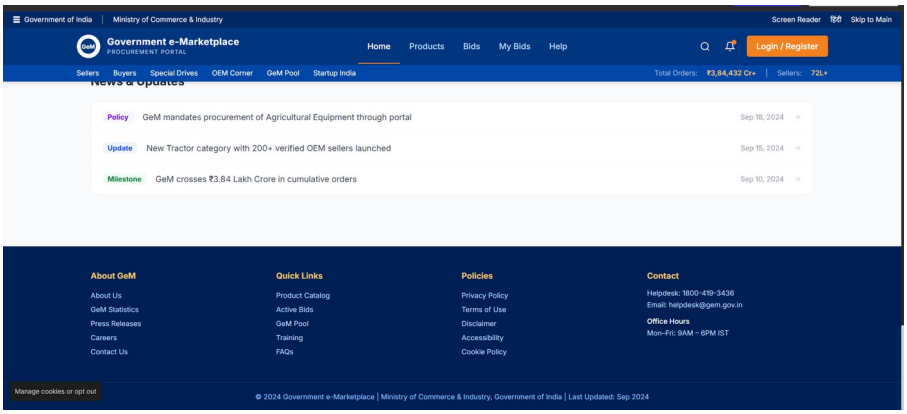
Success is shown by a tick and the words "Bid Submitted Successfully!" and "Bid Confirmed", and the "What Happens Next?" steps are numbered Step 1 to Step 4.

Figure 5: Active Tractor Bids



Each bid card writes the deadline as text with a clock icon and also gives the full closing date and time.

Figure 6: News & Updates and footer



News items carry a text tag (Policy, Update, Milestone) as well as colour, and footer links use light text on dark navy for high contrast.

What Colour Blind Mode Changes

The selector offers Normal, Protanopia, Deuteranopia and Tritanopia options. When a mode is chosen, the changes below apply across the site.

Element	Colour-only risk	Colour Blind Mode change
Closing-time badges (Closes in 2h red, 2d / 11d orange)	Red and orange look alike to protan and deutan users	Keep the clock icon and time text, add the word “Urgent” and a heavier border for bids closing within 24 hours
GeM Verified badge (green)	Green on a light photo can disappear	Keep the tick plus text and add a solid dark outline
Selected states (Tractors tile, radio buttons)	Selection must not rely on hue	Dark fill with a tick mark and a thick border, filled dot for radios
Success banner (Bid Submitted Successfully)	Green banner means “success” only by colour	Tick icon and the heading words stay, with a left border pattern
News tags (Policy, Update, Milestone)	Purple, blue and green tags look similar	Text label plus a distinct icon for each tag
Orange primary buttons (Login / Register, Search)	Orange on blue can lose contrast	Bold text, underline on hover and focus, contrast of at least 4.5:1
Palette in Colour Blind Mode	Red / green pairs	Use the colour-blind-safe Okabe–Ito palette: blue, orange, sky blue, yellow and vermillion

How to Verify

1. Open the site in Chrome, press F12, then open Rendering and choose “Emulate vision deficiencies”.
2. Check protanopia, deuteranopia and tritanopia with Colour Blind Mode off and then on.
3. Confirm that each badge, selection and status can still be told apart by its icon, text or border.